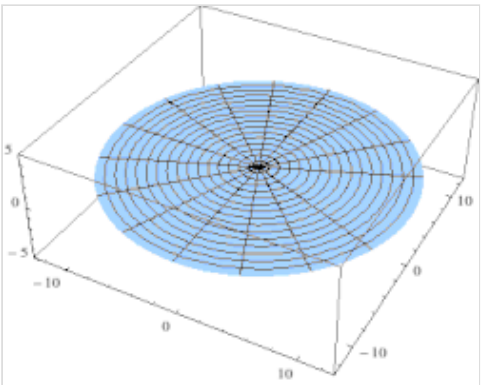
Bessel function

Bessel functions are a class of special functions that commonly appear in problems involving wave motion, heat conduction, and other physical phenomena with circular or cylindrical symmetry. They are named after the German astronomer and mathematician Friedrich Bessel, who studied them systematically in 1824.[1]



Bessel functions describe the radial part of vibrations of a circular membrane.

Bessel functions are solutions to a particular type of ordinary differential equation:

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + \left(x^2 - \alpha^2\right) y = 0,$$

where α is a number that determines the shape of the solution. This number is called the *order* of the Bessel function and can be any complex number. Although the same equation arises for both α and $-\alpha$, mathematicians define separate Bessel functions for each to ensure the functions behave smoothly as the order changes.

The most important cases are when α is an integer or a half-integer. When α is an integer, the resulting Bessel functions are often called **cylinder functions** or **cylindrical harmonics** because they naturally arise when solving problems (like Laplace's equation) in cylindrical coordinates. When α is a half-integer, the solutions are called **spherical Bessel functions** and are used in spherical systems, such as in solving the Helmholtz equation in spherical coordinates.

Applications

Bessel's equation arises when finding separable solutions to Laplace's equation and the Helmholtz equation in cylindrical or spherical coordinates. Bessel functions are therefore especially important for many problems of wave propagation and static potentials. In solving problems in cylindrical coordinate systems, one obtains Bessel functions of integer order ($\alpha = n$); in spherical problems, one obtains half-integer orders ($\alpha = n + 1/2$). For example:

- Electromagnetic waves in a cylindrical waveguide
- Pressure amplitudes of inviscid rotational flows
- Heat conduction in a cylindrical object
- Modes of vibration of a thin circular or annular acoustic membrane (such as a drumhead or other membranophone) or thicker plates such as sheet metal (see Kirchhoff–Love plate theory, Mindlin–Reissner plate theory)
- Diffusion problems on a lattice
- Solutions to the Schrödinger equation in spherical and cylindrical coordinates for a free particle
- Position space representation of the Feynman propagator in quantum field theory
- Solving for patterns of acoustical radiation
- Frequency-dependent friction in circular pipelines
- Dynamics of floating bodies

- Angular resolution
- Diffraction from helical objects, including DNA
- Probability density function of product of two normally distributed random variables[2]
- Analyzing of the surface waves generated by microtremors, in geophysics and seismology.

Bessel functions also appear in other fields, such as signal processing (e.g., see FM audio synthesis, Kaiser window, or Bessel filter).

Definitions

Because this is a linear differential equation, solutions can be scaled to any amplitude. The amplitudes chosen for the functions originate from the early work in which the functions appeared as solutions to definite integrals rather than solutions to differential equations. Because the differential equation is second-order, there must be two linearly independent solutions: one of the first kind and one of the second kind. Depending upon the circumstances, however, various formulations of these solutions are convenient. Different variations are summarized in the table below and described in the following sections.The subscript n is typically used in place of α when α is known to be an integer.

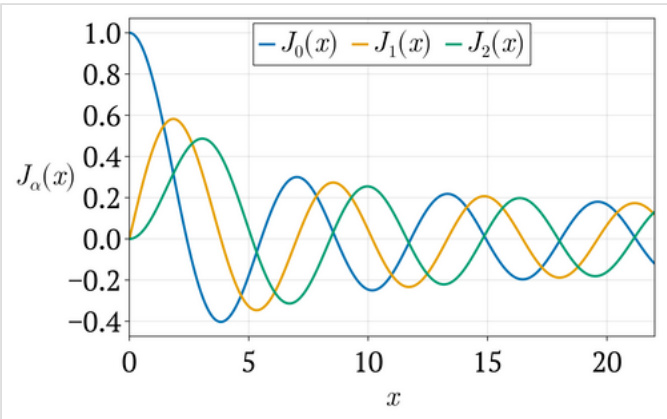
Type	First kind	Second kind
Bessel functions	J_α	Y_α
Modified Bessel functions	I_α	K_α
Hankel functions	$H_\alpha^{(1)} = J_\alpha + iY_\alpha$	$H_\alpha^{(2)} = J_\alpha - iY_\alpha$
Spherical Bessel functions	j_n	y_n
Modified spherical Bessel functions	i_n	k_n
Spherical Hankel functions	$h_n^{(1)} = j_n + iy_n$	$h_n^{(2)} = j_n - iy_n$

Bessel functions of the second kind and the spherical Bessel functions of the second kind are sometimes denoted by N_n and n_n, respectively, rather than Y_n and y_n.[3][4]

Bessel functions of the first kind: J_α

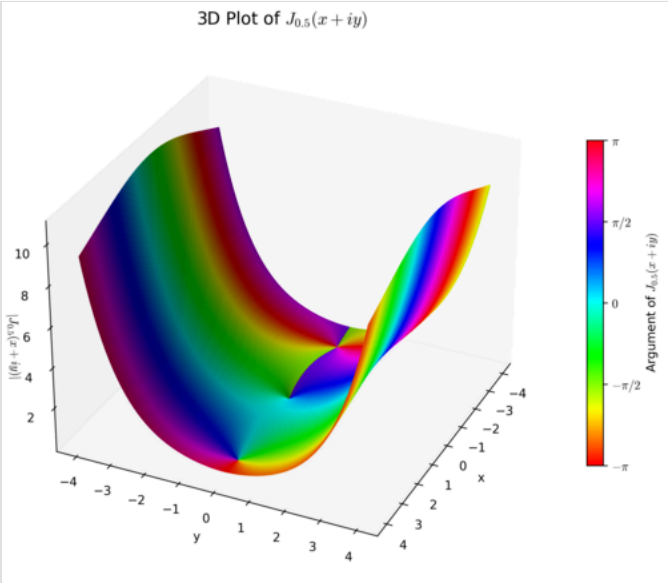
Bessel functions of the first kind, denoted as $J_\alpha(x)$, are solutions of Bessel's differential equation. For integer or positive α, Bessel functions of the first kind are finite at the origin ($x = 0$); while for negative non-integer α, Bessel functions of the first kind diverge as x approaches zero. It is possible to define the function by x^α times a Maclaurin series (note that α need not be an integer, and non-integer powers are not permitted in a Taylor series), which can be found by applying the Frobenius method to Bessel's equation:[5]

$$J_\alpha(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,\Gamma(m+\alpha+1)} \left(\frac{x}{2}\right)^{2m+\alpha},$$



Plot of Bessel function of the first kind, $J_\alpha(x)$, for integer orders $\alpha = 0, 1, 2$.

where $\Gamma(z)$ is the gamma function, a shifted generalization of the factorial function to non-integer values. Some earlier authors define the Bessel function of the first kind differently, essentially without the division by 2 in $x/2$;[6] this definition is not used in this article. The Bessel function of the first kind is an entire function if α is an integer, otherwise it is a multivalued function with singularity at zero. The graphs of Bessel functions look roughly like oscillating sine or cosine functions that decay proportionally to $x^{-1/2}$ (see also their asymptotic forms below), although their roots are not generally periodic, except asymptotically for large x. (The series indicates that $-J_1(x)$ is the derivative of $J_0(x)$, much like $-\sin x$ is the derivative of $\cos x$; more generally, the derivative of $J_n(x)$ can be expressed in terms of $J_{n\pm 1}(x)$ by the identities below.)



Plot of Bessel function of the first kind $J_\alpha(z)$ with $\alpha = 0.5$ in the plane from $-4-4i$ to $4+4i$.

For non-integer α, the functions $J_\alpha(x)$ and $J_{-\alpha}(x)$ are linearly independent, and are therefore the two solutions of the differential equation. On the other hand, for integer order n, the following relationship is valid (the gamma function has simple poles at each of the non-positive integers):[7]

$$J_{-n}(x) = (-1)^n J_n(x).$$

This means that the two solutions are no longer linearly independent. In this case, the second linearly independent solution is then found to be the Bessel function of the second kind, as discussed below.

Bessel's integrals

Another definition of the Bessel function, for integer values of n, is possible using an integral representation:[8]

$$J_n(x) = \frac{1}{\pi}\int_0^\pi \cos(n\tau - x\sin\tau)\,d\tau = \frac{1}{\pi}\operatorname{Re}\left(\int_0^\pi e^{i(n\tau - x\sin\tau)}\,d\tau\right),$$

which is also called Hansen-Bessel formula.[9]

This was the approach that Bessel used,[10] and from this definition he derived several properties of the function. The definition may be extended to non-integer orders by one of Schläfli's integrals, for $\operatorname{Re}(x) > 0$:[8][11][12][13][14]

$$J_\alpha(x) = \frac{1}{\pi}\int_0^\pi \cos(\alpha\tau - x\sin\tau)\,d\tau - \frac{\sin(\alpha\pi)}{\pi}\int_0^\infty e^{-x\sinh t - \alpha t}\,dt.$$

Relation to hypergeometric series

The Bessel functions can be expressed in terms of the generalized hypergeometric series as[15]

$$J_\alpha(x) = \frac{\left(\frac{x}{2}\right)^\alpha}{\Gamma(\alpha+1)}\,{}_0F_1\left(\alpha+1; -\frac{x^2}{4}\right).$$

This expression is related to the development of Bessel functions in terms of the Bessel–Clifford function.

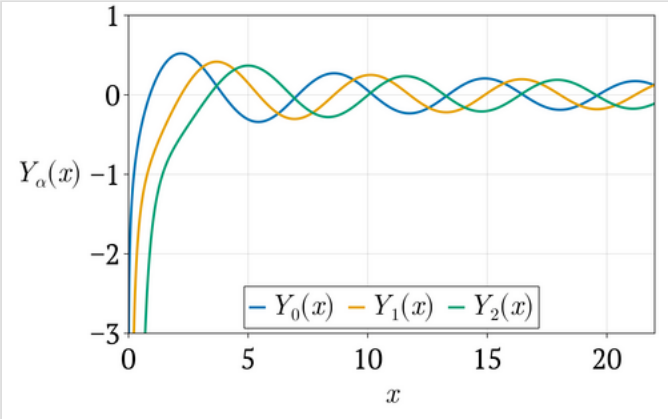
Relation to Laguerre polynomials

In terms of the Laguerre polynomials L_k and arbitrarily chosen parameter t, the Bessel function can be expressed as[16]

$$\frac{J_\alpha(x)}{\left(\frac{x}{2}\right)^\alpha} = \frac{e^{-t}}{\Gamma(\alpha+1)} \sum_{k=0}^{\infty} \frac{L_k^{(\alpha)}\left(\frac{x^2}{4t}\right)}{\binom{k+\alpha}{k}} \frac{t^k}{k!}.$$

Bessel functions of the second kind: Y_α

The Bessel functions of the second kind, denoted by $Y_\alpha(x)$, occasionally denoted instead by $N_\alpha(x)$, are solutions of the Bessel differential equation that have a singularity at the origin ($x = 0$) and are multivalued. These are sometimes called **Weber functions**, as they were introduced by H. M. Weber (1873), and also **Neumann functions** after Carl Neumann.[17]



Plot of Bessel function of the second kind, $Y_\alpha(x)$, for integer orders $\alpha = 0, 1, 2$

For non-integer α, $Y_\alpha(x)$ is related to $J_\alpha(x)$ by

$$Y_\alpha(x) = \frac{J_\alpha(x)\cos(\alpha\pi) - J_{-\alpha}(x)}{\sin(\alpha\pi)}.$$

In the case of integer order n, the function is defined by taking the limit as a non-integer α tends to n:

$$Y_n(x) = \lim_{\alpha\to n} Y_\alpha(x).$$

If n is a nonnegative integer, we have the series[18]

$$Y_n(z) = -\frac{\left(\frac{z}{2}\right)^{-n}}{\pi} \sum_{k=0}^{n-1} \frac{(n-k-1)!}{k!} \left(\frac{z^2}{4}\right)^k + \frac{2}{\pi} J_n(z) \ln\frac{z}{2} - \frac{\left(\frac{z}{2}\right)^n}{\pi} \sum_{k=0}^{\infty} (\psi(k+1) + \psi(n+k+1)$$

where $\psi(z)$ is the digamma function, the logarithmic derivative of the gamma function.[4]

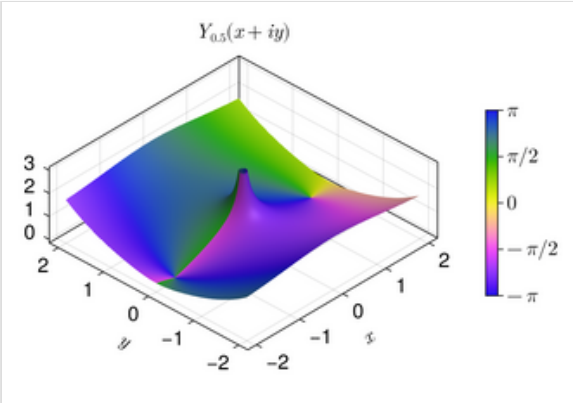
There is also a corresponding integral formula (for $\mathrm{Re}(x) > 0$):[19]

$$Y_n(x) = \frac{1}{\pi}\int_0^\pi \sin(x\sin\theta - n\theta)\,d\theta - \frac{1}{\pi}\int_0^\infty \left(e^{nt} + (-1)^n e^{-nt}\right) e^{-x\sinh t}\,dt.$$

In the case where $n = 0$: (with γ being Euler's constant)

$$Y_0(x) = \frac{4}{\pi^2}\int_0^{\frac{1}{2}\pi} \cos(x\cos\theta)\left(\gamma + \ln\left(2x\sin^2\theta\right)\right)\,d\theta.$$

$Y_\alpha(x)$ is necessary as the second linearly independent solution of the Bessel's equation when α is an integer. But $Y_\alpha(x)$ has more meaning than that. It can be considered as a "natural" partner of $J_\alpha(x)$. See also the subsection on Hankel functions below.



Plot of the Bessel function of the second kind $Y_\alpha(z)$ with $\alpha = 0.5$ in the complex plane from $-2-2i$ to $2+2i$.

When α is an integer, moreover, as was similarly the case for the functions of the first kind, the following relationship is valid:

$$Y_{-n}(x) = (-1)^n Y_n(x).$$

Both $J_\alpha(x)$ and $Y_\alpha(x)$ are holomorphic functions of x on the complex plane cut along the negative real axis. When α is an integer, the Bessel functions J are entire functions of x. If x is held fixed at a non-zero value, then the Bessel functions are entire functions of α.

The Bessel functions of the second kind, when α is an integer, are an example of the second kind of solution in Fuchs's theorem.

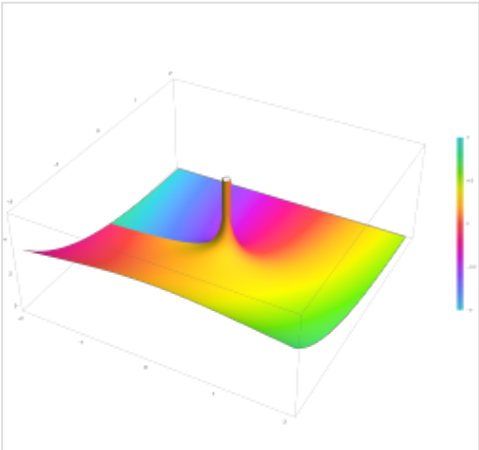
Hankel functions: $H_\alpha^{(1)}$, $H_\alpha^{(2)}$

Another important formulation of the two linearly independent solutions to Bessel's equation are the **Hankel functions of the first and second kind**, $H_\alpha^{(1)}(x)$ and $H_\alpha^{(2)}(x)$, defined as[20]

$$H_\alpha^{(1)}(x) = J_\alpha(x) + iY_\alpha(x),$$

$$H_\alpha^{(2)}(x) = J_\alpha(x) - iY_\alpha(x),$$

where i is the imaginary unit. These linear combinations are also known as **Bessel functions of the third kind**; they are two linearly independent solutions of Bessel's differential equation. They are named after Hermann Hankel.



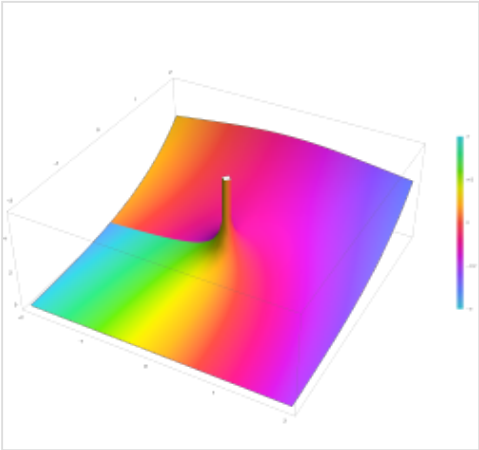
Plot of the Hankel function of the first kind $H_n^{(1)}(x)$ with $n = -0.5$ in the complex plane from $-2-2i$ to $2+2i$

These forms of linear combination satisfy numerous simple-looking properties, like asymptotic formulae or integral representations. Here, "simple" means an appearance of a factor of the form $e^{i\,f(x)}$. For real $x > 0$ where $J_\alpha(x)$, $Y_\alpha(x)$ are real-valued, the Bessel functions of the first and second kind are the real and imaginary parts, respectively, of the first Hankel function and the real and negative imaginary parts of the second Hankel function. Thus, the above formulae are analogs of Euler's formula, substituting $H_\alpha^{(1)}(x)$, $H_\alpha^{(2)}(x)$ for $e^{\pm ix}$ and $J_\alpha(x)$, $Y_\alpha(x)$ for $\cos(x)$, $\sin(x)$, as explicitly shown in the asymptotic expansion.

The Hankel functions are used to express outward- and inward-propagating cylindrical-wave solutions of the cylindrical wave equation, respectively (or vice versa, depending on the sign convention for the frequency).

Using the previous relationships, they can be expressed as

$$H_\alpha^{(1)}(x) = \frac{J_{-\alpha}(x) - e^{-\alpha\pi i} J_\alpha(x)}{i \sin \alpha\pi},$$
$$H_\alpha^{(2)}(x) = \frac{J_{-\alpha}(x) - e^{\alpha\pi i} J_\alpha(x)}{-i \sin \alpha\pi}.$$



Plot of the Hankel function of the second kind $H_n^{(2)}(x)$ with $n = -0.5$ in the complex plane from $-2 - 2i$ to $2 + 2i$

If α is an integer, the limit has to be calculated. The following relationships are valid, whether α is an integer or not:[21]

$$H_{-\alpha}^{(1)}(x) = e^{\alpha\pi i} H_\alpha^{(1)}(x),$$
$$H_{-\alpha}^{(2)}(x) = e^{-\alpha\pi i} H_\alpha^{(2)}(x).$$

In particular, if $\alpha = m + \frac{1}{2}$ with m a nonnegative integer, the above relations imply directly that

$$J_{-(m+\frac{1}{2})}(x) = (-1)^{m+1} Y_{m+\frac{1}{2}}(x),$$
$$Y_{-(m+\frac{1}{2})}(x) = (-1)^m J_{m+\frac{1}{2}}(x).$$

These are useful in developing the spherical Bessel functions (see below).

The Hankel functions admit the following integral representations for $\mathrm{Re}(x) > 0$:[22]

$$H_\alpha^{(1)}(x) = \frac{1}{\pi i} \int_{-\infty}^{+\infty+\pi i} e^{x \sinh t - \alpha t}\, dt,$$
$$H_\alpha^{(2)}(x) = -\frac{1}{\pi i} \int_{-\infty}^{+\infty-\pi i} e^{x \sinh t - \alpha t}\, dt,$$

where the integration limits indicate integration along a contour that can be chosen as follows: from $-\infty$ to 0 along the negative real axis, from 0 to $\pm\pi i$ along the imaginary axis, and from $\pm\pi i$ to $+\infty \pm \pi i$ along a contour parallel to the real axis.[19]

Modified Bessel functions: I_α, K_α

The Bessel functions are valid even for complex arguments x, and an important special case is that of a purely imaginary argument. In this case, the solutions to the Bessel equation are called the **modified Bessel functions** (or occasionally the **hyperbolic Bessel functions**) **of the first and second kind** and are defined as[23]

$$I_\alpha(x) = i^{-\alpha} J_\alpha(ix) = \sum_{m=0}^{\infty} \frac{1}{m!\,\Gamma(m+\alpha+1)} \left(\frac{x}{2}\right)^{2m+\alpha},$$
$$K_\alpha(x) = \frac{\pi}{2} \frac{I_{-\alpha}(x) - I_\alpha(x)}{\sin \alpha\pi},$$

when α is not an integer. When α is an integer, then the limit is used. These are chosen to be real-valued for real and positive arguments x. The series expansion for $I_\alpha(x)$ is thus similar to that for $J_\alpha(x)$, but without the alternating $(-1)^m$ factor.

K_α can be expressed in terms of Hankel functions:

$$K_\alpha(x) = \begin{cases} \frac{\pi}{2} i^{\alpha+1} H_\alpha^{(1)}(ix) & -\pi < \arg x \le \frac{\pi}{2} \\ \frac{\pi}{2} (-i)^{\alpha+1} H_\alpha^{(2)}(-ix) & -\frac{\pi}{2} < \arg x \le \pi \end{cases}$$

Using these two formulae the result to $J_\alpha^2(z) + Y_\alpha^2(z)$, commonly known as Nicholson's integral or Nicholson's formula, can be obtained to give the following

$$J_\alpha^2(x) + Y_\alpha^2(x) = \frac{8}{\pi^2} \int_0^\infty \cosh(2\alpha t) K_0(2x \sinh t)\, dt,$$

given that the condition $\mathrm{Re}(x) > 0$ is met. It can also be shown that

$$J_\alpha^2(x) + Y_\alpha^2(x) = \frac{8\cos(\alpha\pi)}{\pi^2} \int_0^\infty K_{2\alpha}(2x \sinh t)\, dt,$$

only when $|\mathrm{Re}(\alpha)| < \frac{1}{2}$ and $\mathrm{Re}(x) \ge 0$ but not when $x = 0$.[24]

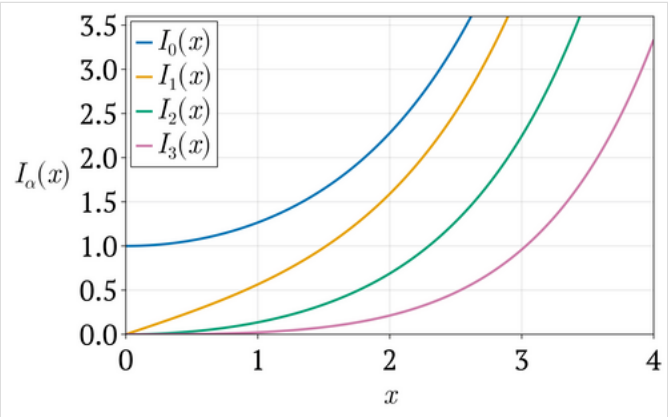
We can express the first and second Bessel functions in terms of the modified Bessel functions (these are valid if $-\pi < \arg z \le \frac{\pi}{2}$):[25]

$$J_\alpha(iz) = e^{\frac{\alpha\pi i}{2}} I_\alpha(z),$$
$$Y_\alpha(iz) = e^{\frac{(\alpha+1)\pi i}{2}} I_\alpha(z) - \frac{2}{\pi} e^{-\frac{\alpha\pi i}{2}} K_\alpha(z).$$

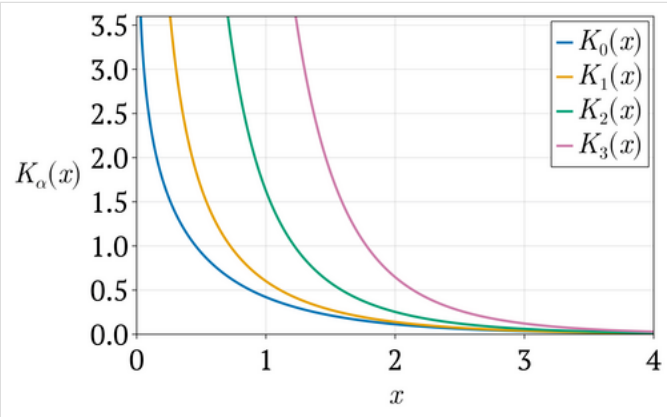
$I_\alpha(x)$ and $K_\alpha(x)$ are the two linearly independent solutions to the **modified Bessel's equation**:[26]

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - \left(x^2 + \alpha^2\right) y = 0.$$

Unlike the ordinary Bessel functions, which are oscillating as functions of a real argument, I_α and K_α are exponentially growing and decaying functions respectively. Like the ordinary Bessel function J_α, the function I_α goes to zero at $x = 0$ for $\alpha > 0$ and is finite at $x = 0$ for $\alpha = 0$. Analogously, K_α diverges at $x = 0$ with the singularity being of logarithmic type for K_0, and $\frac{1}{2}\Gamma(|\alpha|)(2/x)^{|\alpha|}$ otherwise.[27]



Modified Bessel functions of the first kind, $I_\alpha(x)$, for $\alpha = 0, 1, 2, 3$.



Modified Bessel functions of the second kind, $K_\alpha(x)$, for $\alpha = 0, 1, 2, 3$.

Two integral formulas for the modified Bessel functions are (for $\mathrm{Re}(x) > 0$):[28]

$$I_\alpha(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}\cos\alpha\theta\,d\theta - \frac{\sin\alpha\pi}{\pi}\int_0^\infty e^{-x\cosh t-\alpha t}\,dt,$$

$$K_\alpha(x) = \int_0^\infty e^{-x\cosh t}\cosh\alpha t\,dt.$$

Bessel functions can be described as Fourier transforms of powers of quadratic functions. For example (for $\mathrm{Re}(\omega) > 0$):

$$2\,K_0(\omega) = \int_{-\infty}^{\infty}\frac{e^{i\omega t}}{\sqrt{t^2+1}}\,dt.$$

It can be proven by showing equality to the above integral definition for K_0. This is done by integrating a closed curve in the first quadrant of the complex plane.

Modified Bessel functions of the second kind may be represented with Bassett's integral [29]

$$K_n(xz) = \frac{\Gamma\left(n+\frac{1}{2}\right)(2z)^n}{\sqrt{\pi}x^n}\int_0^\infty\frac{\cos(xt)\,dt}{\left(t^2+z^2\right)^{n+\frac{1}{2}}}.$$

Modified Bessel functions $K_{1/3}$ and $K_{2/3}$ can be represented in terms of rapidly convergent integrals[30]

$$K_{\frac{1}{3}}(\xi) = \sqrt{3}\int_0^\infty \exp\left(-\xi\left(1+\frac{4x^2}{3}\right)\sqrt{1+\frac{x^2}{3}}\right)dx,$$

$$K_{\frac{2}{3}}(\xi) = \frac{1}{\sqrt{3}}\int_0^\infty \frac{3+2x^2}{\sqrt{1+\frac{x^2}{3}}}\exp\left(-\xi\left(1+\frac{4x^2}{3}\right)\sqrt{1+\frac{x^2}{3}}\right)dx.$$

The modified Bessel function $K_{\frac{1}{2}}(\xi) = (2\xi/\pi)^{-1/2}\exp(-\xi)$ is useful to represent the Laplace distribution as an Exponential-scale mixture of normal distributions.

The **modified Bessel function of the second kind** has also been called by the following names (now rare):

- **Basset function** after Alfred Barnard Basset
- **Modified Bessel function of the third kind**
- **Modified Hankel function**[31]
- **Macdonald function** after Hector Munro Macdonald

Spherical Bessel functions: j_n, y_n

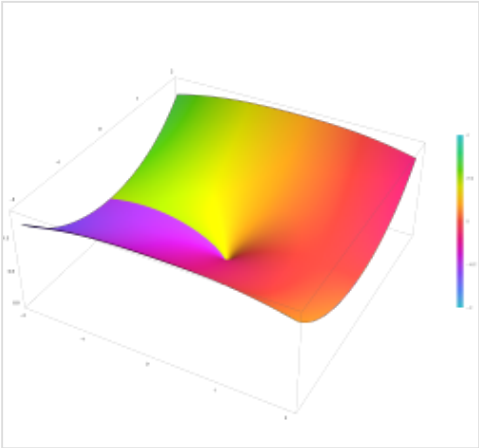
When solving the Helmholtz equation in spherical coordinates by separation of variables, the radial equation has the form

$$x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} + \left(x^2 - n(n+1)\right)y = 0.$$

The two linearly independent solutions to this equation are called the **spherical Bessel functions** j_n and y_n, and are related to the ordinary Bessel functions J_n and Y_n by[32]

$$j_n(x) = \sqrt{\frac{\pi}{2x}} J_{n+\frac{1}{2}}(x),$$
$$y_n(x) = \sqrt{\frac{\pi}{2x}} Y_{n+\frac{1}{2}}(x) = (-1)^{n+1}\sqrt{\frac{\pi}{2x}} J_{-n-\frac{1}{2}}(x).$$

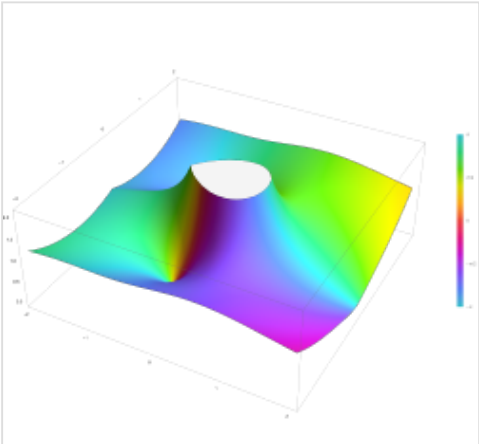
y_n is also denoted n_n or η_n; some authors call these functions the **spherical Neumann functions**.



Plot of the spherical Bessel function of the first kind $j_n(z)$ with $n = 0.5$ in the complex plane from $-2 - 2i$ to $2 + 2i$

From the relations to the ordinary Bessel functions it is directly seen that:

$$j_n(x) = (-1)^n y_{-n-1}(x)$$
$$y_n(x) = (-1)^{n+1} j_{-n-1}(x)$$



Plot of the spherical Bessel function of the second kind $y_n(z)$ with $n = 0.5$ in the complex plane from $-2 - 2i$ to $2 + 2i$

The spherical Bessel functions can also be written as (**Rayleigh's formulas**)[33]

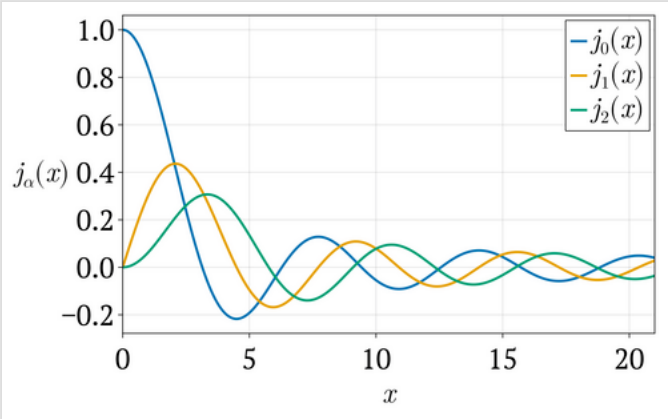
$$j_n(x) = (-x)^n \left(\frac{1}{x}\frac{d}{dx}\right)^n \frac{\sin x}{x},$$
$$y_n(x) = -(-x)^n \left(\frac{1}{x}\frac{d}{dx}\right)^n \frac{\cos x}{x}.$$

The zeroth spherical Bessel function $j_0(x)$ is also known as the (unnormalized) sinc function. The first few spherical Bessel functions are:[34]

$$j_0(x) = \frac{\sin x}{x}.$$
$$j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x},$$
$$j_2(x) = \left(\frac{3}{x^2} - 1\right)\frac{\sin x}{x} - \frac{3\cos x}{x^2},$$
$$j_3(x) = \left(\frac{15}{x^3} - \frac{6}{x}\right)\frac{\sin x}{x} - \left(\frac{15}{x^2} - 1\right)\frac{\mathrm{co}}{}$$

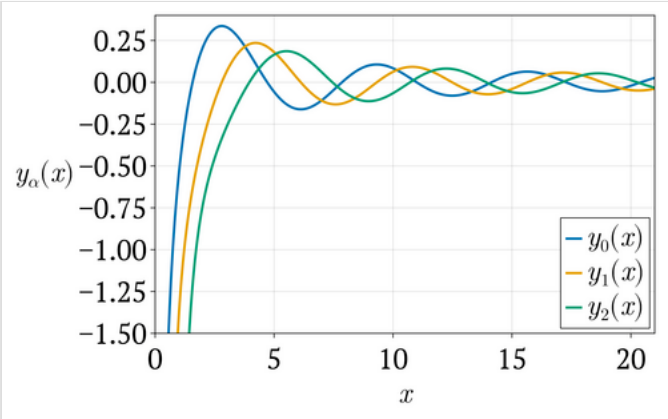
and[35]

$$y_0(x) = -j_{-1}(x) = -\frac{\cos x}{x},$$
$$y_1(x) = j_{-2}(x) = -\frac{\cos x}{x^2} - \frac{\sin x}{x},$$
$$y_2(x) = -j_{-3}(x) = \left(-\frac{3}{x^2} + 1\right)\frac{\cos x}{x} -$$
$$y_3(x) = j_{-4}(x) = \left(-\frac{15}{x^3} + \frac{6}{x}\right)\frac{\cos x}{x} - ($$



Spherical Bessel functions of the first kind $j_\alpha(x)$, for $\alpha = 0, 1, 2$.

The first few non-zero roots of the first few spherical Bessel functions are:



Spherical Bessel functions of the second kind $y_\alpha(x)$, for $\alpha = 0, 1, 2$.

Non-zero Roots of the Spherical Bessel Function (first kind)

Order	Root 1	Root 2	Root 3	Root 4	Root 5
j_0	3.141593	6.283185	9.424778	12.566371	15.707963
j_1	4.493409	7.725252	10.904122	14.066194	17.220755
j_2	5.763459	9.095011	12.322941	15.514603	18.689036
j_3	6.987932	10.417119	13.698023	16.923621	20.121806
j_4	8.182561	11.704907	15.039665	18.301256	21.525418

Non-zero Roots of the Spherical Bessel Function (second kind)

Order	Root 1	Root 2	Root 3	Root 4	Root 5
y_0	1.570796	4.712389	7.853982	10.995574	14.137167
y_1	2.798386	6.121250	9.317866	12.486454	15.644128
y_2	3.959528	7.451610	10.715647	13.921686	17.103359
y_3	5.088498	8.733710	12.067544	15.315390	18.525210
y_4	6.197831	9.982466	13.385287	16.676625	19.916796

Generating function

The spherical Bessel functions have the generating functions[36]

$$\frac{1}{z}\cos\left(\sqrt{z^2-2zt}\right) = \sum_{n=0}^{\infty}\frac{t^n}{n!}j_{n-1}(z),$$
$$\frac{1}{z}\sin\left(\sqrt{z^2-2zt}\right) = \sum_{n=0}^{\infty}\frac{t^n}{n!}y_{n-1}(z).$$

Finite series expansions

In contrast to the whole integer Bessel functions $J_n(x)$, $Y_n(x)$, the spherical Bessel functions $j_n(x)$, $y_n(x)$ have a finite series expression:[37]

$$
\begin{aligned}
j_n(x) &= \sqrt{\frac{\pi}{2x}} J_{n+\frac{1}{2}}(x) \\
&= \frac{1}{2x}\left[e^{ix}\sum_{r=0}^{n}\frac{i^{r-n-1}(n+r)!}{r!(n-r)!(2x)^r} + e^{-ix}\sum_{r=0}^{n}\frac{(-i)^{r-n-1}(n+r)!}{r!(n-r)!(2x)^r}\right] \\
&= \frac{1}{x}\left[\sin\left(x-\frac{n\pi}{2}\right)\sum_{r=0}^{\left\lfloor\frac{n}{2}\right\rfloor}\frac{(-1)^r(n+2r)!}{(2r)!(n-2r)!(2x)^{2r}} + \cos\left(x-\frac{n\pi}{2}\right)\sum_{r=0}^{\left\lfloor\frac{n-1}{2}\right\rfloor}\frac{(-1)^r(n+2r+}{(2r+1)!(n-2r-1}\right.
\end{aligned}
$$

$$
\begin{aligned}
y_n(x) &= (-1)^{n+1} j_{-n-1}(x) = (-1)^{n+1}\frac{\pi}{2x} J_{-\left(n+\frac{1}{2}\right)}(x) \\
&= \frac{(-1)^{n+1}}{2x}\left[e^{ix}\sum_{r=0}^{n}\frac{i^{r+n}(n+r)!}{r!(n-r)!(2x)^r} + e^{-ix}\sum_{r=0}^{n}\frac{(-i)^{r+n}(n+r)!}{r!(n-r)!(2x)^r}\right] \\
&= \frac{(-1)^{n+1}}{x}\left[\cos\left(x+\frac{n\pi}{2}\right)\sum_{r=0}^{\left\lfloor\frac{n}{2}\right\rfloor}\frac{(-1)^r(n+2r)!}{(2r)!(n-2r)!(2x)^{2r}} - \sin\left(x+\frac{n\pi}{2}\right)\sum_{r=0}^{\left\lfloor\frac{n-1}{2}\right\rfloor}\frac{(-1)^r(n}{(2r+1)!(n-}\right.
\end{aligned}
$$

Differential relations

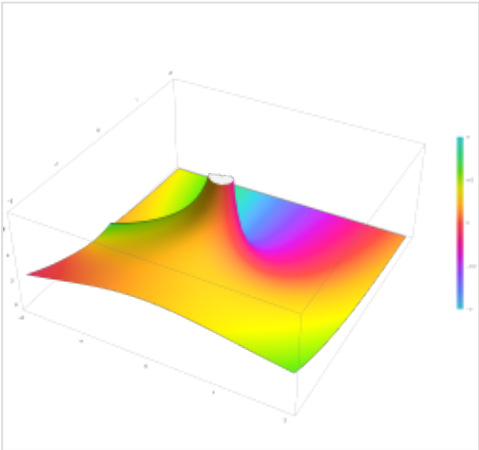
In the following, f_n is any of j_n, y_n, $h_n^{(1)}$, $h_n^{(2)}$ for $n = 0, \pm1, \pm2, \ldots$[38]

$$
\begin{aligned}
\left(\frac{1}{z}\frac{d}{dz}\right)^m \left(z^{n+1} f_n(z)\right) &= z^{n-m+1} f_{n-m}(z), \\
\left(\frac{1}{z}\frac{d}{dz}\right)^m \left(z^{-n} f_n(z)\right) &= (-1)^m z^{-n-m} f_{n+m}(z).
\end{aligned}
$$

Spherical Hankel functions: $h_n^{(1)}$, $h_n^{(2)}$

There are also spherical analogues of the Hankel functions:

$$
\begin{aligned}
h_n^{(1)}(x) &= j_n(x) + i y_n(x), \\
h_n^{(2)}(x) &= j_n(x) - i y_n(x).
\end{aligned}
$$



Plot of the spherical Hankel function of the first kind $h_n^{(1)}(x)$ with $n = -0.5$ in the complex plane from $-2 - 2i$ to $2 + 2i$

There are simple closed-form expressions for the Bessel functions of half-integer order in terms of the standard trigonometric functions, and therefore for the spherical Bessel functions. In particular, for non-negative integers n:

$$
h_n^{(1)}(x) = (-i)^{n+1}\frac{e^{ix}}{x}\sum_{m=0}^{n}\frac{i^m}{m!\,(2x)^m}\frac{(n+m)!}{(n-m)!},
$$

and $h_n^{(2)}$ is the complex-conjugate of this (for real x). It follows, for example, that $j_0(x) = \frac{\sin x}{x}$ and $y_0(x) = -\frac{\cos x}{x}$, and so on.

The spherical Hankel functions appear in problems involving spherical wave propagation, for example in the multipole expansion of the electromagnetic field.

Riccati–Bessel functions: S_n, C_n, ξ_n, ζ_n

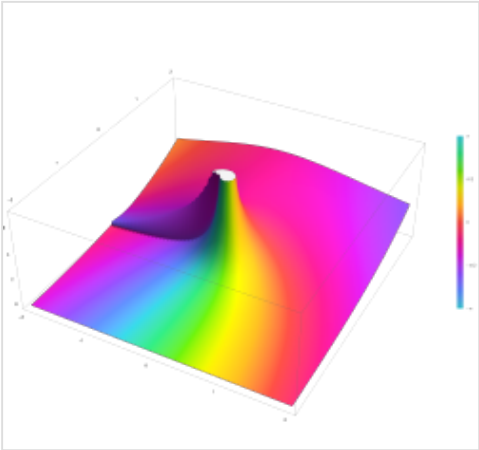
Riccati–Bessel functions only slightly differ from spherical Bessel functions:

$$S_n(x) = x j_n(x) = \sqrt{\frac{\pi x}{2}} J_{n+\frac{1}{2}}(x)$$

$$C_n(x) = -x y_n(x) = -\sqrt{\frac{\pi x}{2}} Y_{n+\frac{1}{2}}(x)$$

$$\xi_n(x) = x h_n^{(1)}(x) = \sqrt{\frac{\pi x}{2}} H^{(1)}_{n+\frac{1}{2}}(x) = S_n(x) - i C_n(x)$$

$$\zeta_n(x) = x h_n^{(2)}(x) = \sqrt{\frac{\pi x}{2}} H^{(2)}_{n+\frac{1}{2}}(x) = S_n(x) + i C_n(x)$$

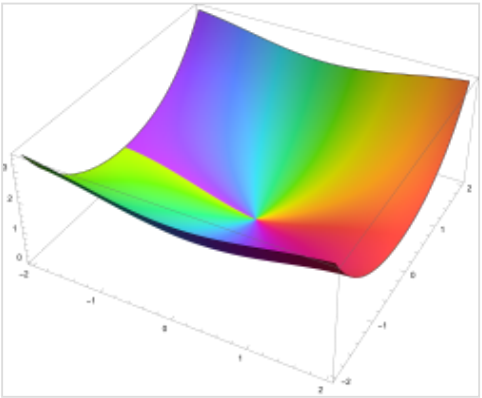


Plot of the spherical Hankel function of the second kind $h_n^{(2)}(x)$ with $n = -0.5$ in the complex plane from $-2 - 2i$ to $2 + 2i$

They satisfy the differential equation

$$x^2 \frac{d^2 y}{dx^2} + \left(x^2 - n(n+1)\right) y = 0.$$

For example, this kind of differential equation appears in quantum mechanics while solving the radial component of the Schrödinger equation with hypothetical cylindrical infinite potential barrier.[39] This differential equation, and the Riccati–Bessel solutions, also arises in the problem of scattering of electromagnetic waves by a sphere, known as Mie scattering after the first published solution by Mie (1908). See e.g., Du (2004)[40] for recent developments and references.



Riccati–Bessel functions Sn complex plot from $-2 - 2i$ to $2 + 2i$

Following Debye (1909), the notation ψ_n, χ_n is sometimes used instead of S_n, C_n.

Asymptotic forms

The Bessel functions have the following asymptotic forms. For small arguments $0 < z \ll \sqrt{\alpha + 1}$, one obtains, when α is not a negative integer:[5]

$$J_\alpha(z) \sim \frac{1}{\Gamma(\alpha+1)} \left(\frac{z}{2}\right)^\alpha.$$

When α is a negative integer, we have

$$J_\alpha(z) \sim \frac{(-1)^\alpha}{(-\alpha)!} \left(\frac{2}{z}\right)^\alpha.$$

For the Bessel function of the second kind we have three cases:

$$Y_\alpha(z) \sim \begin{cases} \frac{2}{\pi}\left(\ln\left(\frac{z}{2}\right)+\gamma\right) & \text{if } \alpha = 0 \\ -\frac{\Gamma(\alpha)}{\pi}\left(\frac{2}{z}\right)^\alpha + \frac{1}{\Gamma(\alpha+1)}\left(\frac{z}{2}\right)^\alpha \cot(\alpha\pi) & \text{if } \alpha \text{ is a positive integer,} \\ -\frac{(-1)^\alpha\Gamma(-\alpha)}{\pi}\left(\frac{z}{2}\right)^\alpha & \text{if } \alpha \text{ is a negative integer,} \end{cases}$$

where γ is the Euler–Mascheroni constant (0.5772...). Note that for the second case (where α is a positive integer) one term will dominate unless α is imaginary.

For large real arguments $z \gg |\alpha^2 - \frac{1}{4}|$, one cannot write a true asymptotic form for Bessel functions of the first and second kind (unless α is half-integer) because they have zeros all the way out to infinity, which would have to be matched exactly by any asymptotic expansion. However, for a given value of $\arg z$ one can write an equation containing a term of order $|z|^{-1}$:[41]

$$J_\alpha(z) = \sqrt{\frac{2}{\pi z}}\left(\cos\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right) + e^{|\mathrm{Im}(z)|}\mathcal{O}\left(|z|^{-1}\right)\right) \qquad \text{for } |\arg z| < \pi,$$

$$Y_\alpha(z) = \sqrt{\frac{2}{\pi z}}\left(\sin\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right) + e^{|\mathrm{Im}(z)|}\mathcal{O}\left(|z|^{-1}\right)\right) \qquad \text{for } |\arg z| < \pi.$$

(For $\alpha = \frac{1}{2}$, the last terms in these formulas drop out completely; see the spherical Bessel functions above.)

The asymptotic forms for the Hankel functions are:

$$H_\alpha^{(1)}(z) \sim \sqrt{\frac{2}{\pi z}} e^{i\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right)} \qquad \text{for } -\pi < \arg z < 2\pi,$$

$$H_\alpha^{(2)}(z) \sim \sqrt{\frac{2}{\pi z}} e^{-i\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right)} \qquad \text{for } -2\pi < \arg z < \pi.$$

These can be extended to other values of $\arg z$ using equations relating $H_\alpha^{(1)}(ze^{im\pi})$ and $H_\alpha^{(2)}(ze^{im\pi})$ to $H_\alpha^{(1)}(z)$ and $H_\alpha^{(2)}(z)$.[42]

It is interesting that although the Bessel function of the first kind is the average of the two Hankel functions, $J_\alpha(z)$ is not asymptotic to the average of these two asymptotic forms when z is negative (because one or the other will not be correct there, depending on the $\arg z$ used). But the asymptotic forms for the Hankel functions permit us to write asymptotic forms for the Bessel functions of first and second kinds for *complex* (non-real) z so long as $|z|$ goes to infinity at a constant phase angle $\arg z$ (using the square root having positive real part):

$$J_\alpha(z) \sim \frac{1}{\sqrt{2\pi z}} e^{i\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right)} \qquad \text{for } -\pi < \arg z < 0,$$

$$J_\alpha(z) \sim \frac{1}{\sqrt{2\pi z}} e^{-i\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right)} \qquad \text{for } 0 < \arg z < \pi,$$

$$Y_\alpha(z) \sim -i\frac{1}{\sqrt{2\pi z}} e^{i\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right)} \qquad \text{for } -\pi < \arg z < 0,$$

$$Y_\alpha(z) \sim i\frac{1}{\sqrt{2\pi z}} e^{-i\left(z - \frac{\alpha\pi}{2} - \frac{\pi}{4}\right)} \qquad \text{for } 0 < \arg z < \pi.$$

For the modified Bessel functions, Hankel developed asymptotic expansions as well:[43][44]

$$I_\alpha(z) \sim \frac{e^z}{\sqrt{2\pi z}} \left(1 - \frac{4\alpha^2 - 1}{8z} + \frac{\left(4\alpha^2 - 1\right)\left(4\alpha^2 - 9\right)}{2!(8z)^2} - \frac{\left(4\alpha^2 - 1\right)\left(4\alpha^2 - 9\right)\left(4\alpha^2 - 25\right)}{3!(8z)^3} + \cdots\right.$$

$$K_\alpha(z) \sim \sqrt{\frac{\pi}{2z}} e^{-z} \left(1 + \frac{4\alpha^2 - 1}{8z} + \frac{\left(4\alpha^2 - 1\right)\left(4\alpha^2 - 9\right)}{2!(8z)^2} + \frac{\left(4\alpha^2 - 1\right)\left(4\alpha^2 - 9\right)\left(4\alpha^2 - 25\right)}{3!(8z)^3} + \right.$$

There is also the asymptotic form (for large real z)[45]

$$I_\alpha(z) = \frac{1}{\sqrt{2\pi z}\sqrt[4]{1 + \frac{\alpha^2}{z^2}}} \exp\left(-\alpha \operatorname{arcsinh}\left(\frac{\alpha}{z}\right) + z\sqrt{1 + \frac{\alpha^2}{z^2}}\right) \left(1 + \mathcal{O}\left(\frac{1}{z\sqrt{1 + \frac{\alpha^2}{z^2}}}\right)\right).$$

When $\alpha = \frac{1}{2}$, all the terms except the first vanish, and we have

$$I_{1/2}(z) = \sqrt{\frac{2}{\pi}} \frac{\sinh(z)}{\sqrt{z}} \sim \frac{e^z}{\sqrt{2\pi z}} \qquad \text{for } |\arg z| < \tfrac{\pi}{2},$$

$$K_{1/2}(z) = \sqrt{\frac{\pi}{2}} \frac{e^{-z}}{\sqrt{z}}.$$

For small arguments $0 < |z| \ll \sqrt{\alpha + 1}$, we have

$$I_\alpha(z) \sim \frac{1}{\Gamma(\alpha + 1)} \left(\frac{z}{2}\right)^\alpha,$$

$$K_\alpha(z) \sim \begin{cases} -\ln\left(\frac{z}{2}\right) - \gamma & \text{if } \alpha = 0 \\ \frac{\Gamma(\alpha)}{2} \left(\frac{2}{z}\right)^\alpha & \text{if } \alpha > 0 \end{cases}$$

Properties

For any Bessel function whose order α is not a negative integer, the derivatives of the function can be defined as: [46]

$$\frac{d}{dx}J_\alpha(x) = J_{\alpha-1}(x) - \frac{\alpha}{x}J_\alpha(x)$$

or, equivalently,

$$\frac{d}{dx}J_\alpha(x) = \frac{\alpha}{x}J_\alpha(x) - J_{\alpha+1}(x)$$

These formulas can be used to determine a recurrence relation for $J_\alpha(x)$, a more general form of which is given below.[46]

For integer order $\alpha = n$, J_n is often defined via a Laurent series for a generating function:

$$e^{\frac{x}{2}\left(t-\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} J_n(x)t^n$$

an approach used by P. A. Hansen in 1843. (This can be generalized to non-integer order by contour integration or other methods.)

Infinite series of Bessel functions in the form $\sum_{\nu=-\infty}^{\infty} J_{N\nu+p}(x)$ where $\nu, p \in \mathbb{Z},\ N \in \mathbb{Z}^+$ arise in many physical systems and are defined in closed form by the Sung series.[47] For example, when N = 3: $\sum_{\nu=-\infty}^{\infty} J_{3\nu+p}(x) = \frac{1}{3}\left[1 + 2\cos\left(x\sqrt{3}/2 - 2\pi p/3\right)\right]$. More generally, the Sung series and the alternating Sung series are written as:

$$\sum_{\nu=-\infty}^{\infty} J_{N\nu+p}(x) = \frac{1}{N}\sum_{q=0}^{N-1} e^{ix\sin 2\pi q/N} e^{-i2\pi pq/N}$$

$$\sum_{\nu=-\infty}^{\infty} (-1)^\nu J_{N\nu+p}(x) = \frac{1}{N}\sum_{q=0}^{N-1} e^{ix\sin (2q+1)\pi/N} e^{-i(2q+1)\pi p/N}$$

A series expansion using Bessel functions (Kapteyn series) is

$$\frac{1}{1-z} = 1 + 2\sum_{n=1}^{\infty} J_n(nz).$$

Another important relation for integer orders is the *Jacobi–Anger expansion*:

$$e^{iz\cos\phi} = \sum_{n=-\infty}^{\infty} i^n J_n(z) e^{in\phi}$$

and

$$e^{iz\sin\theta} \equiv \sum_{n=-\infty}^{\infty} J_n(z)\, e^{in\theta}.$$

The latter is equivalent to

$$e^{\pm iz\sin\phi} = J_0(z) + 2\sum_{n=1}^{\infty} J_{2n}(z)\cos(2n\phi) \pm 2i\sum_{n=0}^{\infty} J_{2n+1}(z)\sin((2n+1)\phi)$$

which is used to expand a plane wave as a sum of cylindrical waves, or to find the Fourier series of a tone-modulated FM signal.

More generally, a series

$$f(z) = a_0^\nu J_\nu(z) + 2 \cdot \sum_{k=1}^{\infty} a_k^\nu J_{\nu+k}(z)$$

is called Neumann expansion of f. The coefficients for $\nu = 0$ have the explicit form

$$a_k^0 = \frac{1}{2\pi i} \int_{|z|=c} f(z) O_k(z)\, dz$$

where O_k is Neumann's polynomial.[48]

Selected functions admit the special representation

$$f(z) = \sum_{k=0}^{\infty} a_k^\nu J_{\nu+2k}(z)$$

with

$$a_k^\nu = 2(\nu + 2k) \int_0^\infty f(z) \frac{J_{\nu+2k}(z)}{z}\, dz$$

due to the orthogonality relation

$$\int_0^\infty J_\alpha(z) J_\beta(z) \frac{dz}{z} = \frac{2}{\pi} \frac{\sin\left(\frac{\pi}{2}(\alpha - \beta)\right)}{\alpha^2 - \beta^2}$$

More generally, if f has a branch-point near the origin of such a nature that

$$f(z) = \sum_{k=0} a_k J_{\nu+k}(z)$$

then

$$\mathcal{L}\left\{\sum_{k=0} a_k J_{\nu+k}\right\}(s) = \frac{1}{\sqrt{1+s^2}} \sum_{k=0} \frac{a_k}{\left(s + \sqrt{1+s^2}\right)^{\nu+k}}$$

or

$$\sum_{k=0} a_k \xi^{\nu+k} = \frac{1+\xi^2}{2\xi} \mathcal{L}\{f\}\left(\frac{1-\xi^2}{2\xi}\right)$$

where $\mathcal{L}\{f\}$ is the Laplace transform of f.[49]

Another way to define the Bessel functions is the Poisson representation formula and the Mehler-Sonine formula:

$$\begin{aligned} J_\nu(z) &= \frac{\left(\frac{z}{2}\right)^\nu}{\Gamma\left(\nu + \frac{1}{2}\right)\sqrt{\pi}} \int_{-1}^{1} e^{izs} \left(1 - s^2\right)^{\nu - \frac{1}{2}}\, ds \\ &= \frac{2}{\left(\frac{z}{2}\right)^\nu \cdot \sqrt{\pi} \cdot \Gamma\left(\frac{1}{2} - \nu\right)} \int_1^\infty \frac{\sin zu}{\left(u^2 - 1\right)^{\nu + \frac{1}{2}}}\, du \end{aligned}$$

where $\nu > -\frac{1}{2}$ and $z \in \mathbf{C}$.[50] This formula is useful especially when working with Fourier transforms.

Because Bessel's equation becomes Hermitian (self-adjoint) if it is divided by x, the solutions must satisfy an orthogonality relationship for appropriate boundary conditions. In particular, it follows that:

$$\int_0^1 x J_\alpha\left(x u_{\alpha,m}\right) J_\alpha\left(x u_{\alpha,n}\right)\, dx = \frac{\delta_{m,n}}{2}\left[J_{\alpha+1}\left(u_{\alpha,m}\right)\right]^2 = \frac{\delta_{m,n}}{2}\left[J'_\alpha\left(u_{\alpha,m}\right)\right]^2$$

where $\alpha > -1$, $\delta_{m,n}$ is the Kronecker delta, and $u_{\alpha,m}$ is the mth zero of $J_\alpha(x)$. This orthogonality relation can then be used to extract the coefficients in the Fourier–Bessel series, where a function is expanded in the basis of the functions $J_\alpha(x\, u_{\alpha,m})$ for fixed α and varying m.

An analogous relationship for the spherical Bessel functions follows immediately:

$$\int_0^1 x^2 j_\alpha\left(x u_{\alpha,m}\right) j_\alpha\left(x u_{\alpha,n}\right)\, dx = \frac{\delta_{m,n}}{2}\left[j_{\alpha+1}\left(u_{\alpha,m}\right)\right]^2$$

If one defines a boxcar function of x that depends on a small parameter ε as:

$$f_\varepsilon(x) = \frac{1}{\varepsilon}\operatorname{rect}\left(\frac{x-1}{\varepsilon}\right)$$

(where rect is the rectangle function) then the Hankel transform of it (of any given order $\alpha > -\frac{1}{2}$), $g_\varepsilon(k)$, approaches $J_\alpha(k)$ as ε approaches zero, for any given k. Conversely, the Hankel transform (of the same order) of $g_\varepsilon(k)$ is $f_\varepsilon(x)$:

$$\int_0^\infty k J_\alpha(kx) g_\varepsilon(k)\, dk = f_\varepsilon(x)$$

which is zero everywhere except near 1. As ε approaches zero, the right-hand side approaches $\delta(x-1)$, where δ is the Dirac delta function. This admits the limit (in the distributional sense):

$$\int_0^\infty k J_\alpha(kx) J_\alpha(k)\, dk = \delta(x-1)$$

A change of variables then yields the *closure equation*:[51]

$$\int_0^\infty x J_\alpha(ux) J_\alpha(vx)\, dx = \frac{1}{u}\delta(u-v)$$

for $\alpha > -\frac{1}{2}$. For the spherical Bessel functions the orthogonality relation is:

$$\int_0^\infty x^2 j_\alpha(ux) j_\alpha(vx)\, dx = \frac{\pi}{2uv}\delta(u-v)$$

for $\alpha > -1$.

Another important property of Bessel's equations, which follows from Abel's identity, involves the Wronskian of the solutions:

$$A_\alpha(x)\frac{dB_\alpha}{dx} - \frac{dA_\alpha}{dx}B_\alpha(x) = \frac{C_\alpha}{x}$$

where A_α and B_α are any two solutions of Bessel's equation, and C_α is a constant independent of x (which depends on α and on the particular Bessel functions considered). In particular,

$$J_\alpha(x)\frac{dY_\alpha}{dx} - \frac{dJ_\alpha}{dx}Y_\alpha(x) = \frac{2}{\pi x}$$

and

$$I_\alpha(x)\frac{dK_\alpha}{dx} - \frac{dI_\alpha}{dx}K_\alpha(x) = -\frac{1}{x},$$

for $\alpha > -1$.

For $\alpha > -1$, the even entire function of genus 1, $x^{-\alpha}J_\alpha(x)$, has only real zeros. Let

$$0 < j_{\alpha,1} < j_{\alpha,2} < \cdots < j_{\alpha,n} < \cdots$$

be all its positive zeros, then

$$J_\alpha(z) = \frac{\left(\frac{z}{2}\right)^\alpha}{\Gamma(\alpha+1)}\prod_{n=1}^{\infty}\left(1 - \frac{z^2}{j_{\alpha,n}^2}\right)$$

(There are a large number of other known integrals and identities that are not reproduced here, but which can be found in the references.)

Recurrence relations

The functions J_α, Y_α, $H_\alpha^{(1)}$, and $H_\alpha^{(2)}$ all satisfy the recurrence relations[52]

$$\frac{2\alpha}{x}Z_\alpha(x) = Z_{\alpha-1}(x) + Z_{\alpha+1}(x)$$

and

$$2\frac{dZ_\alpha(x)}{dx} = Z_{\alpha-1}(x) - Z_{\alpha+1}(x),$$

where Z denotes J, Y, $H^{(1)}$, or $H^{(2)}$. These two identities are often combined, e.g. added or subtracted, to yield various other relations. In this way, for example, one can compute Bessel functions of higher orders (or higher derivatives) given the values at lower orders (or lower derivatives). In particular, it follows that[53]

$$\left(\frac{1}{x}\frac{d}{dx}\right)^m [x^\alpha Z_\alpha(x)] = x^{\alpha-m}Z_{\alpha-m}(x),$$
$$\left(\frac{1}{x}\frac{d}{dx}\right)^m \left[\frac{Z_\alpha(x)}{x^\alpha}\right] = (-1)^m\frac{Z_{\alpha+m}(x)}{x^{\alpha+m}}.$$

Using the previous relations one can arrive to similar relations for the *Spherical* Bessel functions:

$$\frac{2\alpha+1}{x}j_\alpha(x) = j_{\alpha-1} + j_{\alpha+1}$$

and

$$\frac{dj_\alpha(x)}{dx} = j_{\alpha-1} - \frac{\alpha+1}{x} j_\alpha$$

Modified Bessel functions follow similar relations:

$$e^{\left(\frac{x}{2}\right)\left(t+\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} I_n(x) t^n$$

and

$$e^{z\cos\theta} = I_0(z) + 2\sum_{n=1}^{\infty} I_n(z)\cos n\theta$$

and

$$\frac{1}{2\pi}\int_0^{2\pi} e^{z\cos(m\theta)+y\cos\theta} d\theta = I_0(z)I_0(y) + 2\sum_{n=1}^{\infty} I_n(z) I_{mn}(y).$$

The recurrence relation reads

$$C_{\alpha-1}(x) - C_{\alpha+1}(x) = \frac{2\alpha}{x} C_\alpha(x),$$

$$C_{\alpha-1}(x) + C_{\alpha+1}(x) = 2\frac{d}{dx} C_\alpha(x),$$

where C_α denotes I_α or $e^{\alpha i\pi}K_\alpha$. These recurrence relations are useful for discrete diffusion problems.

Transcendence

In 1929, Carl Ludwig Siegel proved that $J_v(x)$, $J'_v(x)$, and the logarithmic derivative $\frac{J'_v(x)}{J_v(x)}$ are transcendental numbers when v is rational and x is algebraic and nonzero.[54] The same proof also implies that $\Gamma(v+1)(2/x)^v J_v(x)$ is transcendental under the same assumptions.[55]

Sums with Bessel functions

The product of two Bessel functions admits the following sum:

$$\sum_{\nu=-\infty}^{\infty} J_\nu(x) J_{n-\nu}(y) = J_n(x+y),$$

$$\sum_{\nu=-\infty}^{\infty} J_\nu(x) J_{\nu+n}(y) = J_n(y-x).$$

From these equalities it follows that

$$\sum_{\nu=-\infty}^{\infty} J_\nu(x) J_{\nu+n}(x) = \delta_{n,0}$$

and as a consequence

$$\sum_{\nu=-\infty}^{\infty} J_\nu^2(x) = 1.$$

These sums can be extended to include a term multiplier that is a polynomial function of the index. For example,

$$\sum_{\nu=-\infty}^{\infty} \nu J_\nu(x) J_{\nu+n}(x) = \frac{x}{2}\left(\delta_{n,1} + \delta_{n,-1}\right),$$

$$\sum_{\nu=-\infty}^{\infty} \nu J_\nu^2(x) = 0,$$

$$\sum_{\nu=-\infty}^{\infty} \nu^2 J_\nu(x) J_{\nu+n}(x) = \frac{x}{2}\left(\delta_{n,-1} - \delta_{n,1}\right) + \frac{x^2}{4}\left(\delta_{n,-2} + 2\delta_{n,0} + \delta_{n,2}\right),$$

$$\sum_{\nu=-\infty}^{\infty} \nu^2 J_\nu^2(x) = \frac{x^2}{2}.$$

Multiplication theorem

The Bessel functions obey a multiplication theorem

$$\lambda^{-\nu} J_\nu(\lambda z) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{\left(1-\lambda^2\right) z}{2}\right)^n J_{\nu+n}(z),$$

where λ and ν may be taken as arbitrary complex numbers.[56][57] For $|\lambda^2 - 1| < 1$,[56] the above expression also holds if J is replaced by Y. The analogous identities for modified Bessel functions and $|\lambda^2 - 1| < 1$ are

$$\lambda^{-\nu} I_\nu(\lambda z) = \sum_{n=0}^{\infty} \frac{1}{n!}\left(\frac{\left(\lambda^2-1\right) z}{2}\right)^n I_{\nu+n}(z)$$

and

$$\lambda^{-\nu} K_\nu(\lambda z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!}\left(\frac{\left(\lambda^2-1\right) z}{2}\right)^n K_{\nu+n}(z).$$

Zeros of the Bessel function

Bourget's hypothesis

Bessel himself originally proved that for nonnegative integers n, the equation $J_n(x) = 0$ has an infinite number of solutions in x.[58] When the functions $J_n(x)$ are plotted on the same graph, though, none of the zeros seem to coincide for different values of n except for the zero at $x = 0$. This phenomenon is known as **Bourget's**

hypothesis after the 19th-century French mathematician who studied Bessel functions. Specifically it states that for any integers $n \geq 0$ and $m \geq 1$, the functions $J_n(x)$ and $J_{n+m}(x)$ have no common zeros other than the one at $x = 0$. The hypothesis was proved by Carl Ludwig Siegel in 1929.[59]

Transcendence

Siegel proved in 1929 that when v is rational, all nonzero roots of $J_v(x)$ and $J'_v(x)$ are transcendental,[60] as are all the roots of $K_v(x)$.[55] It is also known that all roots of the higher derivatives $J_\nu^{(n)}(x)$ for $n \leq 18$ are transcendental, except for the special values $J_1^{(3)}(\pm\sqrt{3}) = 0$ and $J_0^{(4)}(\pm\sqrt{3}) = 0$.[60]

Numerical approaches

For numerical studies about the zeros of the Bessel function, see Gil, Segura & Temme (2007), Kravanja et al. (1998) and Moler (2004).

Numerical values

The first zeros in J_0 (i.e., $j_{0,1}$, $j_{0,2}$ and $j_{0,3}$) occur at arguments of approximately 2.40483, 5.52008 and 8.65373, respectively.[61]

History

Waves and elasticity problems

The first appearance of a Bessel function appears in the work of Daniel Bernoulli in 1732, while working on the analysis of a vibrating string, a problem that was tackled before by his father Johann Bernoulli.[1] Daniel considered a flexible chain suspended from a fixed point above and free at its lower end.[1] The solution of the differential equation led to the introduction of a function that is now considered $J_0(x)$. Bernoulli also developed a method to find the zeros of the function.[1]

Leonhard Euler in 1736, found a link between other functions (now known as Laguerre polynomials) and Bernoulli's solution. Euler also introduced a non-uniform chain that led to the introduction of functions now related to modified Bessel functions $I_n(x)$.[1]

In the middle of the eighteen century, Jean le Rond d'Alembert had found a formula to solve the wave equation. By 1771 there was dispute between Bernoulli, Euler, d'Alembert and Joseph-Louis Lagrange on the nature of the solutions of vibrating strings.[1]

Euler worked in 1778 on buckling, introducing the concept of Euler's critical load. To solve the problem he introduced the series for $J_{\pm 1/3}(x)$.[1] Euler also worked out the solutions of vibrating 2D membranes in cylindrical coordinates in 1780. In order to solve his differential equation he introduced a power series associated to $J_n(x)$, for integer n.[1]

During the end of the 18th century Lagrange, Pierre-Simon Laplace and Marc-Antoine Parseval also found equivalents to the Bessel functions.[1] Parseval, for example, found an integral representation of $J_0(x)$ using cosine.[1]

At the beginning of the 1800s, Joseph Fourier used $J_0(x)$ to solve the heat equation in a problem with cylindrical symmetry.[1] Fourier won a prize of the French Academy of Sciences for this work in 1811.[1] But most of the details of his work, including the use of a Fourier series, remained unpublished until 1822.[1] Poisson, in rivalry with Fourier, extended Fourier's work in 1823, introducing new properties of Bessel functions, including Bessel functions of half-integer order (now known as spherical Bessel functions).[1]

Astronomical problems

In 1770, Lagrange introduced the series expansion of Bessel functions to solve Kepler's equation, a transcendental equation in astronomy. Friedrich Wilhelm Bessel had seen Lagrange's solution but found it difficult to handle. In 1813 in a letter to Carl Friedrich Gauss, Bessel simplified the calculation using trigonometric functions.[1] Bessel published his work in 1819, independently introducing the method of Fourier series unaware of the work of Fourier which was published later.[1] In 1824, Bessel carried out a systematic investigation of the functions, which earned the functions his name.[1] In older literature the functions were called cylindrical functions or even Bessel–Fourier functions.[1]

See also

- Anger function
- Bessel polynomials
- Bessel–Clifford function
- Bessel–Maitland function
- Fourier–Bessel series
- Hahn–Exton *q*-Bessel function
- Hankel transform
- Incomplete Bessel functions
- Jackson *q*-Bessel function
- Kelvin functions
- Kontorovich–Lebedev transform
- Lentz's algorithm
- Lerche–Newberger sum rule
- Lommel function
- Lommel polynomial
- Neumann polynomial
- Schlömilch's series
- Sonine formula
- Struve function
- Vibrations of a circular membrane
- Weber function (defined at Anger function)
- Gauss' circle problem

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